ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success

MOCK TEST

(Academic Session : 2023 - 2024)

TARGET : PRE-MEDICAL 2024

Test Type : **01**   Test Pattern : **NEET (UG)**

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	2	1	2	3	1	1	4	1	4	3	2	3	4	2	3	2	4	1	2	1	4	1	1	1	2	3	2	1	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	3	3	4	1	3	3	3	1	1	3	3	4	1	2	2	4	2	1	2	2	1	1	3	2	2	2	4	2	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	2	1	3	4	4	1	1	2	2	2	4	3	1	1	4	2	4	3	1	1	3	2	3	2	4	3	1	2	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	4	4	2	2	2	4	2	1	3	3	1	1	2	1	2	3	2	3	2	1	2	4	1	1	2	2	2	3	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	1	3	1	4	2	3	4	1	1	1	1	1	1	3	2	1	3	2	2	2	2	2	1	2	2	2	4	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	3	3	4	2	1	1	3	2	3	2	3	4	2	3	2	4	3	3	2	3	3	1	4	1	4	4	1	4	4
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	3	1	3	3	1	3	3	4	3	3	4	4	3	3	3	3	1	2	2	1										

HINT – SHEET

SUBJECT : PHYSICS

SECTION-A

2. Ans (2)

(a) Length (L), mass (M) and velocity (LT^{-1}) are independent.

(b) Pressure ($ML^{-1}T^{-2}$), density (ML^{-3}) and velocity (LT^{-1}) are dependent as

$$\begin{vmatrix} 1 & -1 & -2 \\ 1 & -3 & 0 \\ 0 & 1 & -1 \end{vmatrix} = 1(3-0)+1(-1-0)-2(1-0)=0$$

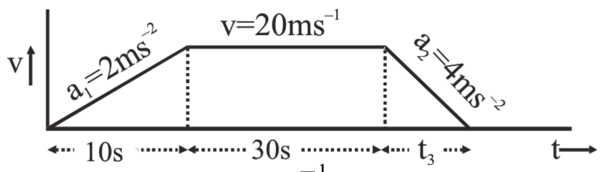
(c) Force (MLT^{-2}), velocity (LT^{-1}) and time (T) are independent as

$$\begin{vmatrix} 1 & 1 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{vmatrix} = 1(1-0)-1(0-0)-2(0-0)=1\neq 0$$

(d) Force $= \dfrac{\text{momentum}}{\text{time}}$

$\Rightarrow$ force, momentum and time are dependent.

3. Ans (1)



$v = a_1t = 2 \times 10 = 20\ ms^{-1}$

also $0 = v - a_2t_3 \quad \Rightarrow t_3 = \dfrac{20}{4} = 5s$

distance = area under the v-t graph

$= \dfrac{1}{2}\times 10\times 2 + 30\times 20 + \dfrac{1}{2}\times 5\times 20$

$= 100 + 600 + 50 = 750$ m

4. Ans (2)

Here final and initial positions are same.

$w_g = 0$

$\Delta k = w_{all} = w_g + w_{air}$

$\dfrac{1}{2}m(v^2-u^2) = 0 + w_{air}$

$\Rightarrow w_{air} = \dfrac{1}{2}\times 1\times(16^2-20^2) = -72$ J

5. Ans (3)

$\vec{v}_2 - \vec{v}_1 = \vec{u}_1 - \vec{u}_2$

$-\hat{i} - \vec{v}_1 = 2\hat{i} - (-\hat{i})$

$\vec{v}_1 = -4\hat{i}$ m/s

6. **Ans (1)**

$T_{max.} = m(r\omega^2 + g)$

$30 = 0.5(2 \times \omega^2 + 10)$

$2\omega^2 = 50$ or $\omega = 5$ rad/sec

7. **Ans (1)**

$T^2 \propto r^3$

$$\frac{T_1}{T_2} = \left(\frac{r_1}{r_2}\right)^{3/2}$$

$$\frac{24}{T_2} = (4)^{3/2} \Rightarrow T_2 = 3\,\text{hrs.}$$

8. **Ans (4)**

$Y = \dfrac{F\ell}{A\Delta\ell}$; In given problem y, ℓ, $\Delta\ell$ are constants

$F \propto \pi r^2$ or $F \propto r^2$

$$\frac{F_1}{F_2} = \frac{r_1^2}{r_2^2} = \frac{1}{4}$$

9. **Ans (1)**

Let the radius of bigger drop is R , smaller drop is r then

$$\frac{4}{3}\pi R^3 = 8 \times \frac{4}{3} \times \pi r^3$$

or $R = 2r$ ...... (i)

Terminal velocity, $v \propto r^2$

$$\therefore \frac{v'}{v} = \frac{R^2}{r^2} = \left(\frac{2r}{r}\right)^2 = 4$$

or $v' = 4v = 4 \times 8 = 32$ cm s^{-1}

10. **Ans (4)**

$10^3 \times \frac{1}{2} \times 10 + 10^3 \times 80 + 1 \times 10^3 \times T$

$= 4.4 \times 10^3 \times 1 \times (30 - T)$

$T = 8.7°$ C

11. **Ans (3)**

$$\frac{Q_1}{t} = \frac{KA(90-60)}{0.6} = 50\ KA$$

and $\dfrac{Q_2}{t} = \dfrac{KA(150-110)}{0.8} = 50\ KA$

12. **Ans (2)**

work done at constant pressure is

$$W = p\Delta V = nR\,\Delta T$$

where, p is pressure, ΔV the volume change, R is the gas constant, ΔT the change is temperature and n is the number of moles.

Given, $n = 1$, $T_2 = 127°C = 400K$

$T_1 = 27°C = 300$ K, $R = 8.31$ J/mol-K

$\therefore$ $W = 1 \times 8.31 \times (400 - 300) \Rightarrow W = 831J$

13. **Ans (3)**

$$K_{eq} = \frac{2k.k}{2k+k} = \frac{2k}{3}$$

$$\therefore T = 2\pi\sqrt{\frac{3m}{2k}}$$

14. **Ans (4)**

$U_{max} = \frac{1}{2}kx^2$

$1 = \frac{1}{2}k(0.2)^2 \Rightarrow k = 50$

and $T = 2\pi\sqrt{\dfrac{m}{k}} \Rightarrow T = 2\pi\sqrt{\dfrac{4}{50}} = \dfrac{2\pi\sqrt{2}}{5}$ sec

15. **Ans (2)**

torque acting on dipole is given by

$\tau = pE\sin\theta$

$20\sqrt{3} = pE\sin 30° = \dfrac{pE}{2}$ or $pE = 40\sqrt{3}$

potential energy of dipole in external electric field $= U = -pE\cos\theta = -40\sqrt{3} \times \dfrac{\sqrt{3}}{2} = -60J$

16. **Ans (3)**

$$\mu = \frac{V_d}{E} = \frac{7.5 \times 10^{-4}}{3 \times 10^{-10}}$$

$\mu = 2.5 \times 10^6$

17. **Ans (2)**

$P = \dfrac{V^2}{R}$ R $\Rightarrow$ constant

$$\frac{P_2}{P_1} = \left(\frac{V_2}{V_1}\right)^2 \Rightarrow P_2 = 500\left(\frac{55}{220}\right)^2$$

$P_2 = 31.25$ watt

18. **Ans (4)**

Given m = 4A-m, $2\ell = 8$ cm, $A = 1$ cm^2

$$I = \frac{M}{V} = \frac{m.2\ell}{A.2\ell} = \frac{m}{A}$$

$$I = \frac{4}{10^{-4}} = 4 \times 10^4 \text{ A/m}$$

19. Ans (1)

$I = I_0(1-e^{-t/\lambda})$

$\lambda = \frac{L}{R} = \frac{10}{5} = 2$

$I_0 = \frac{E_0}{R} = \frac{15}{5} = 3$

$I = 3(1-e^{-2/2})$

$= 3(1-1/e)$

20. Ans (2)

$\frac{V_S}{V_P} = \frac{N_S}{N_P} = 6 \Rightarrow \frac{V_S}{60\sqrt{2}} = 6$

$\Rightarrow V_S = 6 \times 60\sqrt{2} = 360\sqrt{2}$ volt

21. Ans (1)

$v = \frac{C}{\sqrt{\mu_r \varepsilon_r}} = \frac{3 \times 10^8}{\sqrt{1 \times 1.44}}$

$= 2.5 \times 10^8\ ms^{-1}$

22. Ans (4)

Work done in PV graph is Area under the curve.

23. Ans (1)

For grazing at AC

$\frac{3}{2} \sin\theta = \frac{4}{3} \sin 90°$

$\sin\theta = \frac{8}{9}$

for ray to reach BC

$\sin\theta > 8/9$

24. Ans (1)

$m_t = \frac{f}{f-u}$ ....(I)

$m_L = \left(\frac{f}{f-u}\right)^2$ ....(II)

$\frac{m_t}{m_L} = \frac{1}{\left(\frac{f}{f-u}\right)} = \frac{f-u}{f}$

$= \frac{f - \left(\frac{-f}{3}\right)}{f} = \frac{4f}{3f} = \frac{4}{3}$

25. Ans (1)

$\sin\theta = \frac{2\lambda}{a}$

$\theta = \frac{2 \times 6328 \times 10^{-10}}{2 \times 10^{-4}}$

$\theta = 6328 \times 10^{-6}$ rad.

or $\theta = 0.36°$

26. Ans (2)

$\overline{\bar{A} + B + A.B} = (A + B).\left(\overline{A.\bar{B}}\right)$

$= (A + B).\left(\bar{A} + \bar{\bar{B}}\right)$

$= \bar{A}B + \bar{A}B$

27. Ans (3)

When a charged particle (charge q, mass m) enters perpendicularly in a magnetic field (B) then, radius of the path described by it

$r = \frac{mv}{qB} \Rightarrow mv = qBr$

Also de-Broglie wavelength $\lambda = \frac{h}{mv}$

$\Rightarrow \lambda = \frac{h}{qBr} \Rightarrow \frac{\lambda_\alpha}{\lambda_p} = \frac{q_p r_p}{q_\alpha r_\alpha} = \frac{1}{2}$

28. Ans (2)

$f_B = 18/3 = 6$ Hz

$f_2 = f_1 \pm f_B = 341 \pm 6$

29. Ans (1)

$f = \frac{1}{2\pi\sqrt{LC}}$

30. Ans (3)

$\beta = \frac{\lambda D}{d}$

32. Ans (3)

$V = -\frac{GM}{R} - \frac{3GM}{R} = -\frac{4GM}{R}$

33. Ans (3)

$\omega = \frac{2\pi}{4} = \frac{\pi}{2}$

$x = A\sin\omega t$

$x = 5 \sin\frac{\pi}{2}t$

$3 = 5 \sin\left(\frac{\pi}{2}t_1\right)$

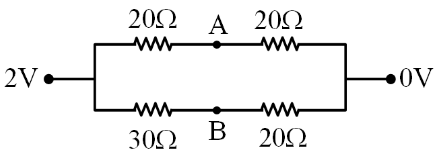
$t_1 = \frac{37}{90}$ sec

$4 = 5 \sin\left(\frac{\pi}{2}t_2\right)$

$t_2 = \frac{53}{90}$ sec

$t = \frac{53}{90} - \frac{37}{90} = \frac{16}{90} = \frac{8}{45}$ sec

34. Ans (4)



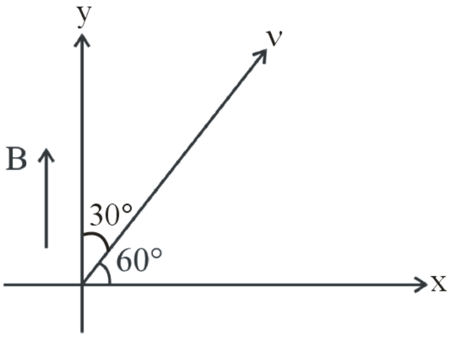
Apply voltage division formula.

35. Ans (1)

$$m = \frac{v_0 - f_0}{f_0}\left(1 + \frac{D}{fe}\right)$$

SECTION-B

36. Ans (3)



Path of proton will be helix

$$r = \frac{mv\sin 30°}{qB}$$

$$r = \frac{1.67 \times 10^{-27} \times 2 \times 10^{6} \times \frac{1}{2}}{1.6 \times 10^{-19} \times 0.104}$$

$r = 0.1$ m

$$T = \frac{2\pi m}{qB} = \frac{2\pi \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.104}$$

$T = 2\pi \times 10^{-7}$ sec.

37. Ans (3)

For maximum range hole must be in the middle of ground and liq. surface.

38. Ans (3)

Theory Base

39. Ans (1)

$$f = \frac{uv}{u - v}$$

$$6 = \frac{-x(25 - x)}{-x - (25 - x)}$$

$$6 = \frac{-25x + x^2}{-x - 25 + x}$$

$x^2 - 25x + 150 = 0$

$x = 10$ cm and 15 cm

40. Ans (1)

$Q_{(i)} = (M_{Br} - M_{Kr})\, c^2 > 0$

$Q_{(ii)} = (M_{Br} - M_{Se} - 2_{Me})\, c^2 > 0$

$Q_{(ii)} = (M_{Br} - M_{Se})\, c^2 > 0$

43. Ans (4)

(a) Concentration of electrons is much higher than concentration of holes so semiconductor is N-type.

(b), (c) If pentavalent impurity is added to pure semiconductor like silicon or germanium, then its electrical conductivity increases and the semiconductor is called N-type semiconductor.

(d) Energy bands in solids are a consequence of Pauli's exclusion principle.

44. Ans (1)

$$K_L = \frac{1}{2}mu^2$$

$$K_H = \frac{1}{2}m\left(\sqrt{u^2 - 2g\ell(1 - \cos 180^\circ)}\right)^2$$

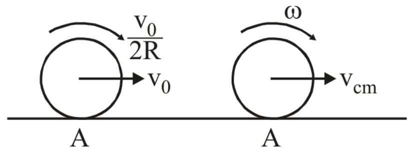
$$K_H = \frac{1}{2}mu^2 - \frac{1}{2}m(4g\ell)$$

$$K_L - K_H = \frac{1}{2}m(4g\ell)$$

$$= \frac{1}{2} \times 1 \times (4 \times 10 \times 1) \text{ J}$$

$= 20$ J

45. Ans (2)



Conservation of angular momentum about A

$\Rightarrow MvR + I\omega$

$$mv_0R + \frac{2}{5}(mR^2)\left(\frac{V_0}{2R}\right)$$

$$= mV_{cm}R + \left(\frac{2}{5}mR^2\right)\left(\frac{V_{cm}}{R}\right)$$

$$V_{cm} = \frac{6V_0}{7}.$$

46. Ans (2)

$I = \frac{5}{4}MR^2 \Rightarrow MR^2 = \frac{4}{5}I$ ......(1)

Now $I' = \frac{3}{2}MR^2$ [from eq(1)]

$= \frac{3}{2} \times \frac{4}{5} I \Rightarrow I' = \frac{6}{5} I$

47. Ans (4)

As x is same

So $T_1 = T_2$

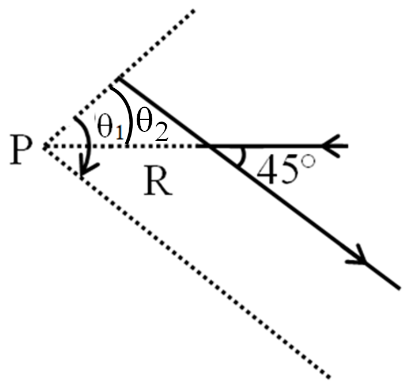
$$\left(\frac{2 \times 3 \times 2}{5}\right) g = \left(\frac{2 \times 4 \times M}{4 + M}\right) g$$

$$\Rightarrow M = \frac{12}{7} kg$$

49. Ans (1)

$\theta_1 = 90°$

$\theta_2 = 45°$



$$B = \frac{\mu_0 I\sqrt{2}}{4\pi R}(\sin 90° - \sin 45°)$$

$$B = \frac{\mu_0 I\sqrt{2}}{\mu\pi R}\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$$

$$B = \frac{\mu_0 I}{4\pi R}(\sqrt{2}-1)$$

50. Ans (2)

$W = CV_2^2 - CV_1^2 = C(225 - 100)$

$W = C(125)$ ....(1)

Now in 2nd case

$W' = C\left[(20)^2 - (15)^2\right] = C\,[400 - 225]$

$W' = C(175)$

(2)/(1)

$$\frac{W'}{W} = \frac{C(175)}{C(125)}$$

$$W' = \frac{7}{5}W$$

SUBJECT : CHEMISTRY

SECTION-A

51. Ans (2)

Chromyl chloride test

52. Ans (1)

S^{2-} test

54. Ans (3)

Spliting energy $(\Delta_0) = [Ni(NH_3)_6]^{2+} >$ $[Ni(H_2O)_6]^{2+}$

55. Ans (2)

No ambidentate ligand

∴ No linkage isomerism

56. Ans (2)

$HNO_2 \xrightarrow{\Delta} HNO_3 + H_2O + NO$

57. Ans (2)

H-bonding in HF

58. Ans (4)

Maximum negative $\Delta_{eg} H = Cl$,

Minimum $\Delta_{eg} H = P$ (due to P^3 configuration)

59. Ans (2)

SF_4 = Seesaw/folded square/distorted tetrahedral

XeF_6 = distorted octahedral

BrF_5 = Square pyramidal

PCl_5 = Trigonal pyramidal

60. Ans (1)

SP hybridisation is called diagonal hybridisation.

61. Ans (3)

In vapour phase of sulphur S_2 molecule has 2 unpaired

63. Ans (1)

$$10 = \frac{x}{x+18} \times 100$$

$10x + 180 = 100x$

$180 = 90x$

$2 = x$

70. Ans (2)

$C_{14}H_{10} + 2H_2O \longrightarrow C_{14}H_8O_2 + 6H^+ + 6e^-$

Moles of $C_{14}H_8O_2 \times 6 = \frac{1 \times 965}{96500} = \frac{1}{100}$

Weight of $C_{14}H_8O_2 = \frac{1}{600} \times 208 = 0.35$ g

71. **Ans (2)**

$5e^- + MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O$

$\Delta E = -\frac{0.0591}{5}\log\frac{1}{[H^+]^8} = -\frac{0.0591}{5}\log\frac{1}{(0.01)^8}$

$= -0.189$ V

72. **Ans (4)**

N_2O_5 has N–O–N bond

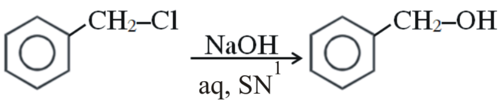
73. **Ans (3)**

(A) is not correct but (R) is correct

74. **Ans (1)**

Lassaigne test used for detection of N, S, P and halogen.

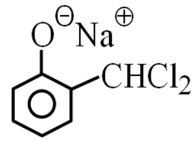
75. **Ans (1)**



76. **Ans (4)**

Conditional

77. **Ans (2)**



is obtained as an intermediate.

78. **Ans (4)**

Ethyl alcohol, Alkan-2-ol (secondary)

Aldehyde & Ketone having $-\overset{\overset{\displaystyle O}{\|}}{C}-CH_3$ group can give Haloform reaction

79. **Ans (3)**

+M stabilise carbocation.

80. **Ans (1)**

$X \longrightarrow CH_3 - CHO$

reagent $\longrightarrow$ NaOH

81. **Ans (1)**

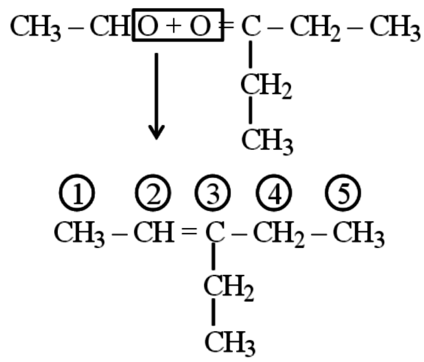
Sucrose doubt shows mutarotation.

83. **Ans (2)**

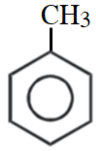
$A \longrightarrow CH_3 - CHO$

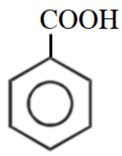
$B \longrightarrow CH_3 - CH = CH - CHO$

84. **Ans (3)**



85. **Ans (2)**

A ⟶ 

B ⟶ 

SECTION-B

87. **Ans (3)**

$CuSO_4(aq) + KCN\ (excess) \longrightarrow K_3[Cu(CN)_4]$

89. **Ans (2)**

Metallic nature P < Si < Be < Mg < Na

95. **Ans (2)**

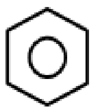
$\pi_1 = 0.05\ RT$

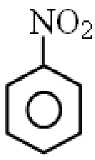
$\pi_2 = 0.05\ RT$

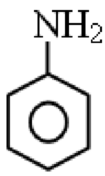
$\pi_3 = \frac{(0.05V + 0.05V)}{2V}RT$

$= 0.05\ RT$

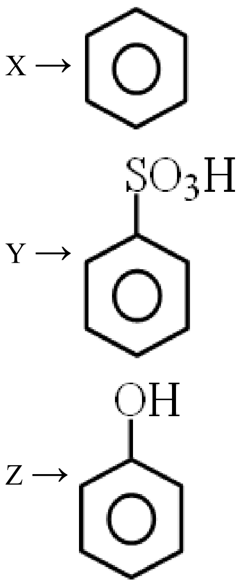
96. **Ans (2)**

X ⟶ 

Y ⟶ 

M & N ⟶ 

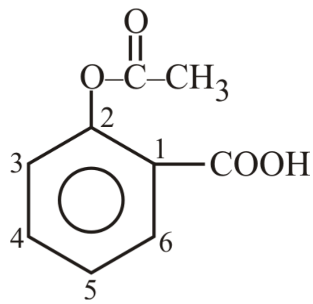
97. **Ans (4)**



98. **Ans (2)**

Steam distillation

99. **Ans (1)**



SUBJECT : BOTANY

SECTION-A

102. **Ans (1)**

NCERT-XI Pg # 40-42

103. **Ans (1)**

NCERT XI, Pg # 35

104. **Ans (2)**

NCERT XIth, Page : 23, Para-2.3.1

106. **Ans (2)**

NCERT XI Page No. # 163

108. **Ans (2)**

NCERT XI Pg. # 247, Para-II

109. **Ans (3)**

NCERT Pg. 91, 93

110. **Ans (2)**

NCERT XI, Pg # 73 & Module

111. **Ans (1)**

NCERT XIth Pg. # 68

119. **Ans (3)**

NCERT XI Pg No. # 20, 21

120. **Ans (1)**

NCERT XIth, Page : 24, Para-2.3.3

121. **Ans (4)**

NCERT Page No. # 167

122. **Ans (2)**

NCERT XIth Pg No. # 128,129

125. **Ans (1)**

NCERT XI Pg.No. 159

126. **Ans (4)**

NCERT XII Pg.No.87, 88, 89

127. **Ans (2)**

NCERT XII Pg.# 116,117

129. **Ans (4)**

NCERT-XII, Page # 89

131. **Ans (1)**

NCERT XII Pg.No. 261 & 262 (15.1.2)

135. **Ans (1)**

NCERT-XII, Pg. No. 112(E), 121(H)

SECTION-B

137. **Ans (2)**

NCERT XIth Page No. # 164,165 and 166

138. **Ans (1)**

NCERT XI Pg # 207, 208,

139. **Ans (3)**

Sesbania, Lupin, Sunhemp

140. **Ans (2)**

NCERT, Pg#183

144. **Ans (2)**

NCERT-XI Pg # 31, 36, 38

145. Ans (1)
NCERT–XI, Pg#135

146. Ans (2)
NCERT Pg.#248, Para-V

147. Ans (2)
NCERT XI, Page : 92 & 96

SUBJECT : ZOOLOGY

SECTION-A

151. Ans (2)
NCERT Pg.# 49, 50, 51

152. Ans (3)
NCERT Pg.# 47

153. Ans (3)
NCERT Pg.# 51

154. Ans (4)
NCERT Pg.# 58

155. Ans (2)
NCERT Pg.# 55

156. Ans (1)
NCERT Pg. # 101

157. Ans (1)
NCERT Pg. # 103

163. Ans (4)
NCERT-XI, Pg No. 263

164. Ans (2)
NCERT–XI, Pg#279,280 18.1.2

165. Ans (3)
NCERT–XI, Pg#284

166. Ans (2)
NCERT-XI, [E] Pg#319, [H] Pg#319

168. Ans (3)
NCERT Pg.# 332

169. Ans (3)
Module 8 Pg.# 73

170. Ans (2)
NCERT XIth Pg.#335

171. Ans (3)
NCERT XII Page # 46

172. Ans (3)
NCERT XII Page # 47

174. Ans (4)
NCERT(XII) Pg#54/51(H) Para: 3.7

180. Ans (4)
NCERT XII (E) Pg. 138

182. Ans (1)
NCERT Pg # 146

185. Ans (1)
NCERT-XII, Pg. # (E)-208, (H)-226

SECTION-B

186. Ans (3)
NCERT Pg.# 49,51,53

188. Ans (4)
NCERT XI Pg.No.268

192. Ans (4)
NCERT Pg.# 332

193. Ans (3)
NCERT Pg.# 333

194. Ans (3)
NCERT XII Page # 43

196. Ans (3)
NCERT-XII, Pg#61

197. Ans (1)
NCERT XII (Bio)_Pg-153

199. Ans (2)
NCERT XII, Page No. # 197, Figure-11.2